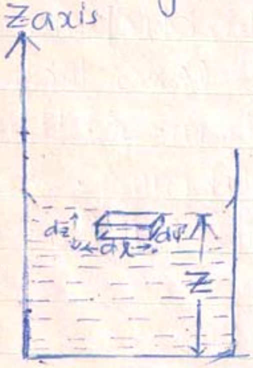
Excess Pressure On a curved liquid surface by using the principle of virtual work:-

Statement: If a system of particles is in equilibrium and if the system receives or is imagined to receive a virtual displacement, the work done due to the virtual displacement is zero and hence the corresponding change in P.E of the system is zero i.e the potential energy of the system, in the position of equilibrium is minimum.

Application of principle of virtual work to the equilibrium of a homogeneous liquid in contact with solid:



Given:

ρ = density of the liquid.

S_1 = Surface tension of the liquid

The total energy of the system Consists of four parts:-

(1). The g.P.E of the liquid:

let us consider an elementary volume $dx\,dy\,dz$ of the liquid on the free surface.

The g.P.E of that element of liquid $= dx\,dy\,dz\,\rho g z$

Hence the total g.P.E of the liquid $= \iiint dx\,dy\,dz\,\rho g z \rightarrow (1)$

(2). The free surface energy of the liquid-air surface $= S_1 \times A \rightarrow (2)$ Where A is the area of the

~~free~~ liquid surface in contact with air.

3) The free surface energy of the liquid-solid surface in contact $= S_2 \times B \longrightarrow (3)$

Where B is the area of solid-liquid surface. & S_2 is the relevent force per unit length acting on the liquid surface in contact with solid.

(4). The free surface energy of the solid-air surface in contact $= S_3 \times C \longrightarrow (4)$

Where C is area of the solid-air surface & S_3 is the relevent force per unit length, acting on the boundary of the surface.

Hence the total P.E of the system:-

$$V = \iiint \rho g z \, dx\,dy\,dz + S_1 A + S_2 B + S_3 C \longrightarrow (5)$$

Since the system is in equilibrium P.E of the system is minimum & if we assume the system it undergo a small virtual displacement; the change in P.E of the system is zero.

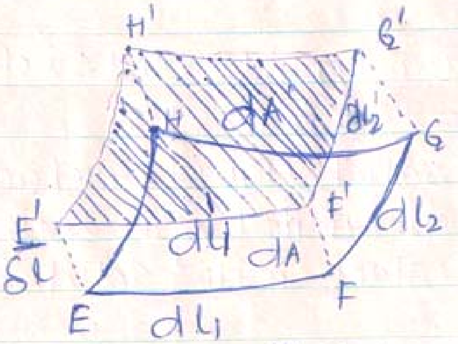
$$\delta V = \delta \left[\iiint \rho g z \, dx\,dy\,dz\right] + S_1 \delta A + S_2 \delta B + S_3 \delta C = 0 \longrightarrow (6)$$

Where δ stands for the change.

Let us consider an element EFGH of area 'dA' on the liquid air surface.

let us assume the element undergo a small virtual displacement 'δl' in a direction, normal to its surface at every point.

$E'F'G'H' = dA'$ is the final position of the element after the virtual displacement.



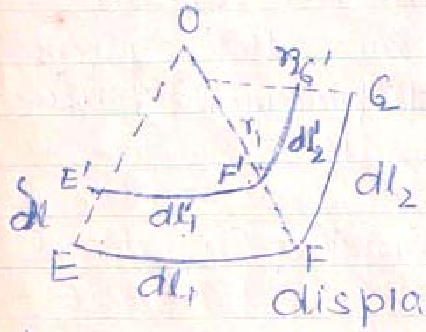
① Change in G.P.E :-

Change in volume of the liquid due to the virtual displacement $= dA\,\delta l$

Change in G.P.E due to the virtual displacement
$= (dA\,\delta l)\rho g z$

Total change in G.P.E of the liquid ; due to the virtual displacement $= \iint \rho g z\, dA\,\delta l \longrightarrow (7)$

② To find dA :- let dl_1 & dl_2 = length of the sides EF & FG of the elementary rectangle EFGH respectively.



dl_1' & dl_2' = the lengths of the corresponding sides E'F' & F'G' after the virtual displacement

let r_1 & r_2 be the radius of curvature of the sides EF & FG respectively

$EE' = FF' = \delta l$ = the arbitrary virtual displacement

From the similar triangles $\dfrac{dl_1'}{dl_1} = \dfrac{r_1 - \delta l}{r_1} = \left(1 - \dfrac{\delta l}{r_1}\right)$

or $dl_1' = dl_1\left(1 - \dfrac{\delta l}{r_1}\right)$

$$\frac{dl_2'}{dl_2} = \frac{r_2 - \delta l}{r_2} = \left(1 - \frac{\delta l}{r_2}\right)$$

Or $dl_2' = dl_2\left(1 - \frac{\delta l}{r_2}\right)$

$\therefore$ The area of the element, after the virtual displacement

$$dA' = dl_1' dl_2' = dl_1 dl_2 \left(1 - \frac{\delta l}{r_1}\right)\left(1 - \frac{\delta l}{r_2}\right)$$

$$\therefore dA' = dA\left[1 - \delta l\left(\frac{1}{r_1} + \frac{1}{r_2}\right) + \frac{(\delta l)^2}{r_1 r_2}\right]$$

δl being very small; its square can be neglected compared to one.

$$\text{or } dA' = dA\left[1 - \delta l\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\right]$$

$\therefore$ The change in area of the element in the liquid air surface $= \delta(dA) = dA' - dA$

$$= dA\left[\cancel{1} - \delta l\left(\frac{1}{r_1} + \frac{1}{r_2}\right) - \cancel{1}\right]$$

$$\delta(dA) = -\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\delta l . dA \longrightarrow (8)$$

Hence the total change in area of the liquid-air surface can be obtained by integrating equation (8).

$$\therefore \delta A = \iint \delta(dA) = \iint -\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\delta l\, dA \longrightarrow (9)$$

To find δB and δC: let us assume for the time being that the line of contact of the liquid free surface with the wall of the container does not shift due to the virtual displacement :-

The change in area of the solid-liquid surface & solid air

Surface will then be zero.

$\therefore \quad \delta B = \delta C = 0 \quad \text{------} \quad (10)$

Putting equation (7), (9) and (10) in equation (6)

$$\delta v = \iint \rho g z \, dA \, \delta l + S_1 \iint -\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\delta l \, dA + S_2 \times 0 + S_3 \times 0 = 0$$

or $\iint \rho g z \, \delta l \, dA - \iint S_1 \left(\frac{1}{r_1}+\frac{1}{r_2}\right)\delta l \, dA = 0 \longrightarrow (11)$

The change in volume of the whole liquid as the liquid-air surface undergoes a virtual displacement

$= \iint \delta l \, dA = 0 \longrightarrow (12)$

Multiplying equation (12) by a constant K and adding with equation (11):

$$\iint \left[(\rho g z + K) - S_1\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\right]\delta l \, dA = 0 \longrightarrow (13)$$

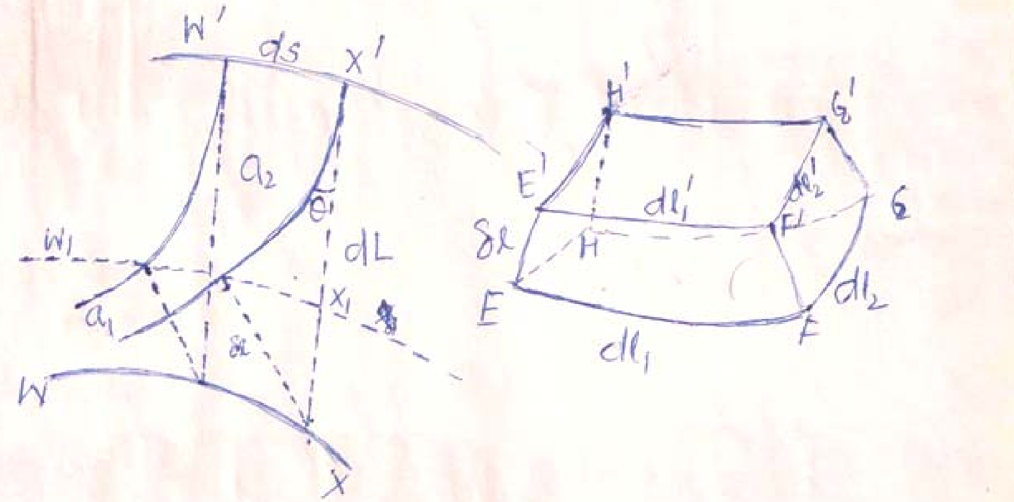
Since the virtual displacement 'δl' is completely arbitrary; the equation (13) will be satisfied only if the bracketed term is zero.

ie $(\rho g z + K) - S_1\left(\frac{1}{r_1}+\frac{1}{r_2}\right) = 0$

$$\rho g z + K = S_1\left(\frac{1}{r_1}+\frac{1}{r_2}\right) \longrightarrow (14)$$

The L.H.S of equation (14) gives the expression for excess pressure.

We now consider the line of contact of the liquid-air surface in contact with the solid wall to shift:



Let WX represent an element on the initial line of contact before the virtual displacement.

W'X' = The element in the line of contact after the virtual displacement.

δL = XX' = The distance through which line of contact shifts along the surface of the wall of the container

Let the normals drawn from the initial line of contact on the liquid surface, cuts the final liquid air surface at $W_1 X_1$

$\delta l = X X_1$ = The $\perp^r$ distance through which the initial liquid-air surface undergoes the displacement.

A_1 = The final area of the liquid air surface after the virtual displacement.

a_1 = The area of the final liquid-air surface which is bounded by the normals drawn to the line of contact of the initial surface.

a_2 = Remaining area of the final liquid-air surface which is obtained from the projection of the area of the wall of the container on the liquid air surface.

θ = angle of contact.

$A_1 = a_1 + a_2$

$$\therefore \delta A = A_1 - A = (a_1 + a_2) - A = (a_1 - A) + a_2 \longrightarrow (15)$$

Consider an element '$w'x'$' of length ds on the final area of contact.

Area of the wall bounded by that element $= ds\,\delta L$

The projection of the area, on the final liquid-air surface after the virtual displacement $= ds.\delta L\cos\theta$

Hence the area of final liquid-air surface between the wall and the normals drawn from the initial line of contact on the final surface

$$a_2 = \iint ds\, dL\cos\theta \longrightarrow (16)$$

$$a_1 - A = -\iint \left(\frac{1}{r_1} + \frac{1}{r_2}\right)\delta L\, dA \longrightarrow (8)$$

[because a_1 is the area after the virtual displacement; considering the displacement to be along the normal to the initial surface as was in the previous case]

$$\delta B = \iint ds\, dL \longrightarrow (17)$$

$$\delta C = -\iint ds\, dL \longrightarrow (18)$$

Using these values; from equation (6) :-

$$\delta V = \iint \rho g z\, \delta l\, dA + S_1 \left[\iint ds.dL\cos\theta + \iint -\left(\frac{1}{r_1}+\frac{1}{r_2}\right).\delta l \ldots\right.$$

$$+ S_2 \iint ds.dL + S_3 \iint -ds\, dL$$

$$\text{or } \delta V = \iint \left[\rho g z - S_1\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\right]\delta l.dA + \iint (S_1\cos\theta + S_2 - S_3)\, ds\, dL = 0 \longrightarrow (19)$$

Since the total change in volume of the liquid due to the virtual displacement is zero.

$$\iint \delta l\, dA = 0 \longrightarrow (20)$$

multiplying equation (20) by a constant K_1 and adding with equation (19) :-

$$\iint \left[(\rho g z + K_1) - S_1\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\right]\delta l.dA + \iint (S_1\cos\theta + S_2 - S_3)\, ds.dL = 0 \longrightarrow (21)$$

Since in the 1st term, the integration extends over the entire liquid-air surface and δl is purely arbitrary, hence the bracketed term in the 1st integration must vanish.

$$\therefore \rho g z + K_1 = S_1\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

$$\text{or} \quad P + P_A = S_1\left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \text{Excess pressure} \longrightarrow (22)$$

Since δL is also arbitrary; the second integration will hold good if the bracked term vanishes ie

$$S_1 \cos\theta_1 + S_2 - S_3 = 0$$

$$\text{or} \quad \cos\theta = \frac{S_3 - S_2}{S_1} \longrightarrow (23)$$

Since S_1, S_2 & S_3 are constants, the angle of contact, between a given liquid and solid surface is a constant.